
TRACEVEIL

AI-Assisted Real-Time Smart Home IoT Digital Forensics and Incident Response Platform

Literature Review

Group 18 — Ayra Jha (K028), Atharva Rangnekar (K053), Aarya Rao (K054) | Faculty Mentor: Prof. Rohit Suryawanshi

1. Papers Studied for This Project

The following papers, datasets, and standards were studied while designing Traceveil. Each entry lists the source, a short explanation of its relevance, and a direct link for verification.

1. TON_IoT Telemetry Dataset: A New Generation Dataset of IoT and IIoT for Data-Driven Intrusion Detection Systems

A. Alsaedi, N. Moustafa, Z. Tari, A. Mahmood, A. Anwar — *IEEE Access*, vol. 8, 2020

Provides heterogeneous IoT/IIoT telemetry, OS logs, and network traffic collected from a realistic UNSW Canberra Cyber Range testbed, labelled normal/attack. We use TON_IoT device telemetry as one of Traceveil's primary public batch-evidence sources for testing validation, normalization, and deterministic investigation workflows.

Link: research.unsw.edu.au/projects/toniot-datasets

2. CASAS Smart Home Datasets

Washington State University — CASAS Project

Provides real smart-home sensor observations — including motion, doors, environmental sensors, timestamps, and activity annotations — collected from instrumented residences. Traceveil uses the Milan smart-home dataset as a primary public source for validating normalization of heterogeneous household sensor events into its Canonical Event Model.

Link: casas.wsu.edu/datasets

3. CICIoT2023: A Real-Time Dataset and Benchmark for Large-Scale Attacks in IoT Environment

E. C. P. Neto, S. Dadkhah, R. Ferreira, A. Zohourian, R. Lu, A. A. Ghorbani — *Sensors*, vol. 23, no. 13, 2023

Traffic captured from 105 real IoT devices across 33 attacks in 7 categories (DDoS, DoS, Recon, Web, Brute Force, Spoofing, Mirai) in a smart-home-like topology. Retained as a secondary network-flow compatibility source for evaluating how Traceveil handles heterogeneous IoT security evidence; it is not part of the primary smart-home evaluation set.

Link: doi.org/10.3390/s23135941

4. IoT-CAD: A Comprehensive Digital Forensics Dataset for AI-based Cyberattack Attribution Detection Methods in IoT Environments

H. Mohamed, N. Koroniotis, F. Schiliro, N. Moustafa — *Ad Hoc Networks*, vol. 174, 2025

The most recent and closest published work to Traceveil's ambition — combines memory, disk, process, system-call and network traces across real/simulated IoT devices and evaluates them with both centralized and federated deep learning for detection and attribution. Studied to understand how forensic datasets are now being paired with AI, and to identify what such approaches still leave out (determinism, investigator-facing correlation).

Link: doi.org/10.1016/j.adhoc.2025.103840

5. Research on Digital Forensics Analyzing Heterogeneous Internet of Things Incident Investigations

D.-H. Shin et al. — *Applied Sciences (MDPI)*, vol. 14, no. 3, 2024

Validates a forensic acquisition methodology across Samsung SmartThings, Aqara, QNAP NAS and Hikvision IP cameras, confirming that proprietary formats and drifted timestamps are a real, device-level obstacle. Studied to ground our 'siloed and fragmented data' problem statement in concrete, tested evidence rather than assumption.

Link: doi.org/10.3390/app14031128

6. Smart Home IoT Forensics: Current Status, Challenges, and Future Directions

Conference paper — 2023 International Conference on Advancement in Computation & Computer Technologies (InCACCT)

Surveys the current state of smart-home IoT forensics and catalogues open challenges: data volatility, webcam/user-data handling, and the lack of standard extraction tooling. Used to scope what is realistically 'in scope' vs. 'out of scope' for an academic prototype.

Link: researchgate.net (InCACCT 2023)

7. A Survey of Internet of Things (IoT) Forensics Frameworks and Challenges

F. Abdel-Fattah et al. — 2023

Reviews existing IoT forensic frameworks, including a blockchain-based evidence-preservation scheme for smart homes, and validates a framework on Samsung SmartThings/Xiaomi Mi Home testbeds. Studied to compare acquisition-and-classification approaches against our normalization-and-correlation approach.

Link: researchgate.net (Abdel-Fattah et al., 2023)

8. Device Forensics in Smart Homes: Insights on Advances, Challenges and Future Directions

Systematic Literature Review — Transactions on Large-Scale Data- and Knowledge-Centered Systems (TLDKS), Springer, 2024

A formal SLR across Springer, IEEE Xplore, ACM Digital Library and ScienceDirect that builds a taxonomy of smart speakers, wearables, and smart-things forensics, and documents manual/logical/physical extraction methods. Used to confirm that we did not find an existing survey proposing a unifying canonical schema — the specific gap Traceveil targets.

Link: researchgate.net (TLDKS, 2024)

9. Cybersecurity and Cyber Forensics for Smart Cities: A Comprehensive Literature Review and Survey

Journal article (154-paper review, 2015–2022) — PMC, 2023

Reviews 154 papers on smart-city cybersecurity and forensics, finding that smart homes/IoT are the most active research area in cybersecurity, while cyber forensics research lags behind at a macro level. Used to support the motivation that forensics — not just detection — needs more research investment in the smart-home space.

Link: ncbi.nlm.nih.gov/pmc/articles/PMC10099346

10. A Common Architecture-Based Smart Home Tools and Applications Forensics for Scalable Investigations

S. Kim, G. Lee, J. Song, I. Lee, T. Shon — Computers, Materials & Continua, vol. 83, no. 1, pp. 661-683, 2025

Proposes a generalized smart-home architecture and a three-stage forensic framework (infer relevant data from application functions → extract/process data from interconnected devices → identify investigation-relevant data), validated on Samsung SmartThings and Xiaomi Mi Home testbeds. Studied as a direct comparison point against Traceveil's own attempt to normalize heterogeneous sources into a common, investigator-facing representation.

Link: aurora.ajou.ac.kr (Kim et al., 2025)

11. Large Language Models in Digital Forensics: Capabilities, Challenges and Future Directions

Review article — ScienceDirect, Forensic Science International: Digital Investigation, 2026

Surveys how LLMs are being used across digital forensics, including LLM-invocation logging for forensic readiness (Chernyshev et al., 2024) and RAG-based timeline summarisation. Studied to justify the design of our AI layer and to identify what 'AI-assisted' should and should not mean in a forensic tool.

Link: sciencedirect.com (LLMs in DF review)

12. ForensicLLM: A Local Large Language Model for Digital Forensics

Journal article — Forensic Science International: Digital Investigation, 2025

Fine-tunes a local LLaMA-3.1-8B model specifically for digital-forensics question answering, and reviews prior work (Scanlon et al. 2023 on ChatGPT's forensic limitations; Michelet & Breitingner 2023 on local vs. cloud LLM report generation). Informed our decision to keep AI inference local rather than cloud-hosted, for both confidentiality and reliability reasons.

Link: sciencedirect.com (ForensicLLM, 2025)

13. NIST SP 800-86 — Guide to Integrating Forensic Techniques into Incident Response

National Institute of Standards and Technology

Defines a four-stage forensic process of Collection → Examination → Analysis → Reporting. Traceveil adapts this general forensic workflow for heterogeneous IoT evidence by introducing explicit validation and canonical normalization before deterministic analysis.

Link: csrc.nist.gov/pubs/sp/800/86/final

14. NIST SP 800-92 — Guide to Computer Security Log Management

National Institute of Standards and Technology

Defines log management, retention, and normalization best practice. Used as the standards basis for our canonical event model and evidence-store design.

Link: csrc.nist.gov/pubs/sp/800/92/final

2. Detailed Literature Review

2.1 IoT and Smart-Home Forensic Frameworks

The foundation of Traceveil's problem statement rests on a well-documented finding across [5], [6], [7], [8], and [10]: smart-home forensic investigation is obstructed by device heterogeneity, not by a lack of general forensic theory. Shin et al. [5] demonstrate this concretely by validating their acquisition methodology across four different vendor ecosystems — Samsung SmartThings, Aqara, QNAP NAS, and Hikvision — and showing that each requires a different extraction approach because of proprietary storage formats and clock drift. The InCACCT 2023 review [6] and the TLDKS systematic literature review [8] both independently confirm this at a survey level: the SLR in particular screens four major databases (Springer, IEEE Xplore, ACM Digital Library, ScienceDirect) and still concludes that the field organizes itself around device categories (smart speakers, wearables, smart-things) rather than a shared data model. Abdel-Fattah et al. [7] propose a framework-based methodology validated on Samsung SmartThings/Xiaomi Mi Home testbeds, and Kim et al. [10] propose a generalized common-architecture approach with a three-stage forensic framework aimed explicitly at scalable investigation across multiple smart-home platforms, also validated on Samsung SmartThings/Xiaomi Mi Home. Together these works show the field converging on acquisition-and-classification, and increasingly on common-architecture normalization, as the units of contribution.

Taken together, this cluster establishes two things for our project: first, that the fragmentation problem stated on our Review-1 slide is not an assumption but a repeatedly observed finding in current literature; and second, that among the frameworks reviewed in this study, we did not identify one that normalizes both historical dataset evidence and controlled live-device telemetry into a single provenance-preserving canonical event model — which is exactly the architectural gap Traceveil's Canonical Event Model is designed to close.

2.2 Benchmark Datasets for IoT Security and Digital Forensics

Traceveil's primary public evaluation sources are CASAS smart-home telemetry [2] and TON_IoT device telemetry [1], supplemented by controlled simulation. CASAS [2] provides household sensor activity — motion, door, and environmental observations — collected from real instrumented residences (we use the Milan dataset specifically), while TON_IoT [1] provides smart-device telemetry containing both normal and attack-labelled behaviour from a realistic testbed. CICIOT2023 [3] is retained as a secondary network-flow compatibility source rather than a primary smart-home dataset; its testbed of 105 real devices is nonetheless useful for evaluating how Traceveil's normalization pipeline handles evidence outside its primary smart-home domain.

IoT-CAD [4], published in 2025 — after our Review-1 slide was prepared — is the most directly comparable prior work we found. It combines memory, disk, process, system-call, and network traces from real and simulated IoT devices, and evaluates them using both centralized and federated deep learning for cyberattack detection and attribution. This is the closest existing attempt at combining 'digital forensics' and 'AI' in an IoT context. However, its evaluation is entirely classifier-driven (accuracy/attribution scores from deep learning models), not investigator-facing — it does not produce a reproducible timeline, a canonical cross-device event model, or an explainable, rule-based risk score. Studying IoT-CAD sharpened our own positioning: Traceveil is not competing to build a better attack classifier, but to build the investigator-facing correlation and timeline layer that sits on top of exactly this kind of evidence.

2.3 AI- and LLM-Assisted Digital Forensics

This cluster [11], [12] directly shaped the design of Traceveil's AI layer. The review of LLMs in digital forensics [11] surveys work such as Chernyshev et al. (2024) on structured LLM-invocation logging for forensic readiness, and RAG-

based timeline summarisation research, showing that natural-language interfaces are becoming a genuine capability for investigators to query forensic data conversationally. ForensicLLM [12] goes further by reviewing Scanlon et al.'s (2023) empirical finding that ChatGPT is useful for regex generation, keyword search, and summarisation, but is prone to hallucination and non-deterministic output when asked to establish forensic facts — and Michelet & Breitingner's (2023) comparison of cloud (ChatGPT-3.5) versus local (LLaMA-2-13B) models for forensic report generation, which found the cloud model more accurate but the local model preferable for confidentiality.

These findings support Traceveil's decision to keep AI inference local rather than relying on an external cloud service, so case evidence does not need to leave the machine; Qwen 9B through Ollama is the implementation selected for the current prototype. Separately, and more fundamentally, the AI layer is restricted by design to explaining results that the deterministic Python engine has already computed — it is never permitted to calculate a risk score, correlation, or detection verdict itself. This is stated explicitly in our architecture as the 'Forensic Authority Mandate' and is a direct, literature-backed response to the hallucination risk documented in [11] and [12].

2.4 Standards

NIST SP 800-86 [13] describes a forensic process centred on Collection, Examination, Analysis, and Reporting. Traceveil follows the same overall evidentiary progression while adding explicit Validate and Normalize stages to deal with heterogeneous IoT sources before deterministic analysis. NIST SP 800-92 [14] further informs the project's approach to structured log handling and consistent event representation, underpinning the design of our canonical event model and evidence store.

3. What Makes Traceveil Different from Existing Solutions in the Market

The table below compares Traceveil against the two categories of solutions already available in the market — enterprise IoT security platforms (Microsoft Defender for IoT, AWS IoT Device Defender) and general-purpose SIEM/investigation platforms (Splunk Enterprise Security, IBM QRadar) — capability by capability.

Capability	Enterprise IoT Platforms	SIEM / Investigation Platforms	Traceveil (Ours)
Live IoT Monitoring	Enterprise-grade (Microsoft Defender for IoT, AWS IoT Device Defender)	Splunk ES / QRadar: general security-event monitoring, not IoT-specific	Controlled laboratory telemetry purpose-built for academic smart-home experiments
Historical Evidence Handling	Product- and deployment-specific; typically centred around their own monitoring ecosystem rather than heterogeneous offline forensic datasets	Strong log investigation, but log format is generic, not IoT-canonical	Core project focus — historical dataset evidence is a first-class input, not an afterthought
Event Normalization	Platform-specific schemas (SmartThings, Xiaomi, Hikvision each different)	General security schemas (CEF/LEEF), not IoT-device aware	One canonical IoT event model shared by every source, live or historical
Timeline Reconstruction	Broader IT/OT workflows, not IoT-forensic specific	Investigation capability exists but is manual/analyst-driven	Core project focus — automatic, reproducible timeline reconstruction
Cross-Device Correlation	Enterprise/fleet-level correlation across many identical devices	Broad event correlation across arbitrary log sources	IoT-specific correlation tuned to smart-home entity relationships
AI-Assisted Investigation	Platform-specific copilots (vendor lock-in)	Product-dependent AI add-ons, usually cloud-based	Planned local, evidence-grounded Qwen 9B assistant — explains only, never computes forensic facts
Data Confidentiality	Deployment-dependent; cloud-connected and on-premises configurations exist depending on the product	Deployment-dependent; both cloud-hosted and on-premises architectures exist	Designed for local processing, with the optional AI model hosted locally and no requirement to transmit case evidence to an external AI API
Reproducibility of Findings	Product-dependent; proprietary scoring may limit transparency or exact reproducibility	Depends on configured rules, analytics, and product implementation	Versioned deterministic rules and risk scoring; identical validated inputs and configuration are designed to reproduce the same forensic outputs
Primary Scope	Enterprise IoT/OT fleets	Enterprise SOC/SIEM at organisational scale	Academic smart-home IoT DFIR prototype — deliberately scoped, not a claimed enterprise replacement

Research Positioning: Traceveil is not built to replace Microsoft Defender for IoT, AWS IoT Device Defender, Splunk ES, or IBM QRadar at enterprise scale. Its contribution is narrower and specific — a single canonical event model that unifies batch evidence and controlled live IoT telemetry, a deterministic and reproducible forensic pipeline, and a locally-hosted AI layer that explains findings without ever computing them. Among the commercial platforms and academic systems reviewed for this project, we did not identify a system that combines all three properties in the

same form: a common batch/live canonical event model, deterministic forensic analysis, and a locally hosted evidence-grounded AI explanation layer.